GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

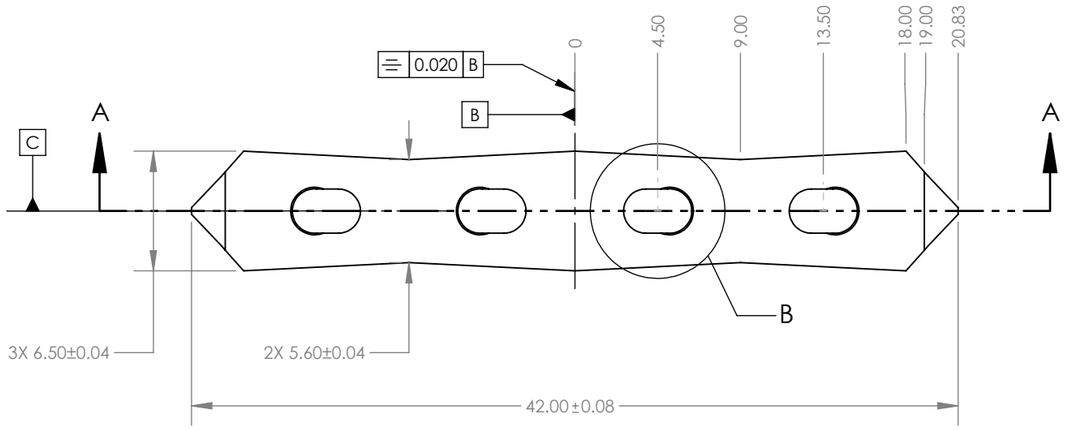
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-ORTHO-IMPLANT-LCP-5.2.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/- 0.03 mm, ANGULAR +/- 0.2 DEG (FINE PRECISION IMPLANT MACHINING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

2. MATERIAL & BIOCOMPATIBILITY SPECIFICATIONS:
2.1 MATERIAL: MEDICAL GRADE TITANIUM ALLOY Ti-6Al-4V ELI (EXTRA LOW INTERSTITIAL) PER ASTM F136 / ISO 5832-3.
2.2 RAW MATERIAL MUST HAVE 100% ULTRASONIC FLAW INSPECTION, FULL CHEMICAL ANALYSIS, AND LOT TRACEABILITY PER ISO 13485.
2.3 100% HOMOGENEOUS SUBTRACTIVE HIGH-SPEED 5-AXIS CNC MACHINING ONLY. NO ADDITIVE MANUFACTURING (SLM/EBM) OR POWDER METALLURGY PERMITTED TO ELIMINATE RISK OF BIOINCOMPATIBLE MICRO-PARTICULATE SHEDDING.

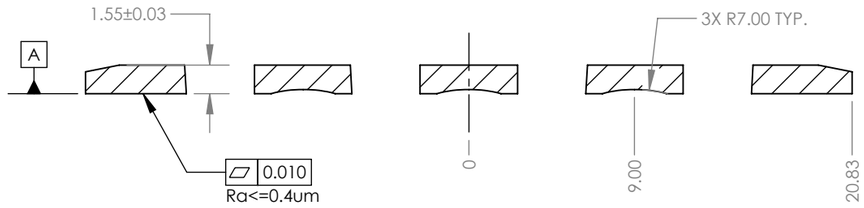
3. IMPLANT GEOMETRY & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (BONE-FACING SURFACE): FLATNESS <= 0.010 mm, SURFACE FINISH Ra <= 0.4 MICRONS.
3.2 SECONDARY DATUM [B] (CENTRAL FRACTURE SYMMETRY LINE): SYMMETRY WITHIN 0.020 mm RELATIVE TO CONTOUR.
3.3 COMBI-HOLES (4X): TRUE POSITION WITHIN 0.020 mm RELATIVE TO [A][B][C]. DCU SLOTS MUST ORIENT CENTRALLY TOWARD FRACTURE DATUM [B] TO ENSURE SYMMETRIC AXIAL DYNAMIC COMPRESSION.
3.4 LIMITED-CONTACT UNDERSURFACE: 3X TRANSVERSE FLUTES (DEPTH 0.22 mm, R 7.00 mm) TO PRESERVE PERIOSTEAL BLOOD SUPPLY.
3.5 LOW-PROFILE CONTOURING: UPPER DORSAL CROWN AND LATERAL EDGES MUST MAINTAIN CONTINUOUS BLEND (R >= 0.40 mm) TO PREVENT TENDON/SOFT TISSUE IRRITATION.

4. DEBURRING & SURFACE QUALITY:
4.1 100% OPTICAL INSPECTION UNDER 30X MICROSCOPY. ABSOLUTELY ZERO RESIDUAL BURRS, CHIPS, CHATTER, OR SHARP EDGES PERMITTED.
4.2 SURFACE FINISHING: CONTROLLED ELECTRO-POLISHING FOLLOWED BY TYPE II MEDICAL ANODIZATION PER ASTM F86 (FATIGUE RESISTANCE ENHANCEMENT).

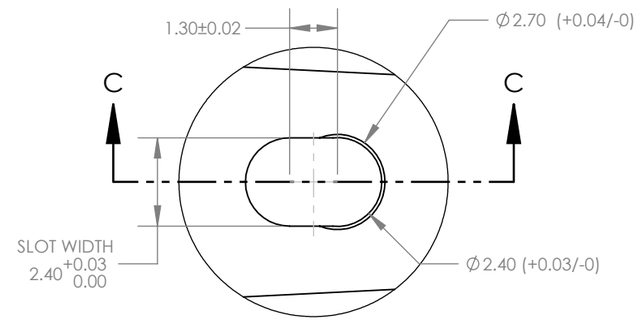
5. CLEANLINESS & PACKAGING:
5.1 COMPLIANT WITH ISO 14644-1 CLASS 4 CLEANROOM STANDARDS.
5.2 MULTI-STAGE ULTRASONIC CAVITATION WASH IN PURIFIED DEIONIZED WATER FOLLOWED BY AIR-TIGHT MEDICAL IMPLANT PACKAGING.



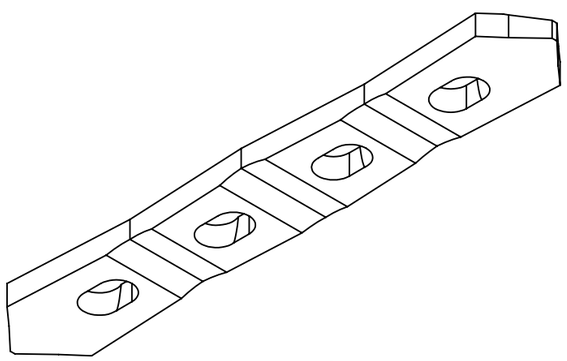
NOTE: DCU SLOTS DIRECTED CENTRALLY TOWARD FRACTURE DATUM [B]



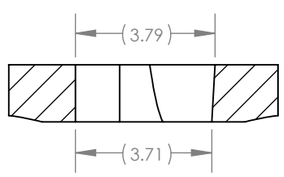
SECTION A-A



DETAIL B



FOR REFERENCE



SECTION C-C

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.03 mm Angular ±0.2°	FINISH: Electropolished & Anodized Type II (ASTM F86), Ultrasonic Cleaned		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE		TITLE: ANATOMICAL LOW-PROFILE LOCKING COMPRESSION PLATE (4-HOLE MINI LCP)	
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Medical Grade Titanium Alloy Ti-6Al-4V ELI (ASTM F136 / ISO 5832-3)	DWG NO. W3-ORTHO-IMPLANT-LCP-5.2	A3
				WEIGHT: ~1.45 g	SCALE: NTS	SHEET 1 OF 1